

REVIEW ARTICLE

Intraoperative hypotension and adverse postoperative outcomes: a systematic review and meta-analysis of randomised controlled trials

Marc Sicova^{1,2}, Jamal Alkadri^{3,4} , Daniel Sibley^{1,2}, Jimmy Schenk⁵ , Matthijs Kant⁶, Teus Kappen⁶ , Denise P. Veelo⁵ , Stuart A. McCluskey^{1,7}, Daniel Santa Mina^{1,2}, Ian Randall^{7,8}  and Wilton A. van Klei^{1,6,7,*} 

¹Department of Anesthesia and Pain Management, Toronto General Hospital – University Health Network, Toronto, ON, Canada, ²Faculty of Kinesiology and Physical Education, University of Toronto, Toronto, ON, Canada, ³Department of Anesthesiology and Pain Medicine, University of Ottawa, Ottawa, ON, Canada, ⁴Division of Cardiac Anesthesiology, University of Ottawa Heart Institute, Ottawa, ON, Canada, ⁵Department of Anesthesiology, Intensive Care and Medicine, University Medical Center Amsterdam, Amsterdam, the Netherlands, ⁶Department of Anesthesiology, Intensive Care and Emergency Medicine, University Medical Center Utrecht, Utrecht, the Netherlands, ⁷Department of Anesthesiology and Pain Medicine, University of Toronto, Toronto, ON, Canada and ⁸Department of Anaesthesia and Pain Management, Toronto Western Hospital – University Health Network, Toronto, ON, Canada

*Corresponding author. E-mail: wilton.vanklei@uhn.ca

Abstract

Background: Intraoperative arterial hypotension commonly occurs and is associated with adverse outcomes. Evidence on causality from RCTs is sparse. This meta-analysis aimed to summarise evidence from RCTs that categorise groups based on MAP thresholds, examining the impact on mortality, acute kidney injury (AKI), and myocardial injury in patients undergoing noncardiac surgery.

Methods: A systematic search of PubMed, Embase, and Web of Science was conducted for RCTs in adults undergoing noncardiac surgery that categorised groups based on MAP thresholds. The primary objective was to examine the effect of differences in achieved MAP on the primary outcome of all-cause mortality within 30 days of surgery. AKI, myocardial injury, and MAP achieved in the intervention and control arms were analysed as secondary outcomes. For each study, two groups were identified: an intervention ('higher-pressure' target) group, and a control ('lower-pressure' target) group. Exploratory analysis examined whether the difference in achieved MAP between groups modified the study outcomes.

Results: Nine RCTs comprising 14 658 patients did not demonstrate a difference in the primary outcome of all-cause mortality between intervention and control groups (RR: 1.06, 95% CI: 0.77–1.45, $I^2=0\%$). No differences were observed for secondary outcomes in 13 RCTs. Differences in intraoperative MAP between study groups were not associated with study outcomes.

Conclusions: Higher vs lower intraoperative MAP did not affect all-cause 30-day mortality, AKI, or myocardial injury; however, only limited differences in blood pressure were achieved between study groups. Stratification by differences in achieved pressure did not alter the results.

Systematic review protocol. PROSPERO (CRD420251230069).

Keywords: acute kidney injury; intraoperative hypotension; mortality; myocardial injury; noncardiac surgery; postoperative outcomes

Editor's key points

- Intraoperative hypotension is associated with adverse outcomes, though evidence on causality from randomised trials of blood pressure targets is minimal.
- The authors conducted a meta-analysis of 9 RCTs (14 658 patients) undergoing noncardiac surgery, where the effect of differences in achieved MAP in these trials was examined with a primary outcome of all-cause mortality within 30 days of surgery. AKI, myocardial injury, and MAP achieved in the intervention and control arms were analysed as secondary outcomes.
- The fixed-effect model reported no difference in all-cause mortality, and trial sequential analysis showed insufficient evidence to conclude there is no effect if a 20% relative risk reduction or lower is considered clinically relevant. In addition, there were no differences in AKI, myocardial injury, and MAP achieved. There was no association of intraoperative MAP and any of the outcomes.
- No differences in clinically relevant and patient-centred outcomes were seen in this meta-analysis of achieved MAP in intraoperative blood pressure target trials. The lack of significant separation in the blood pressure of target groups could be one reason, which remains a challenge in interventional trials of this nature in the intraoperative period.

Haemodynamic instability is commonly observed in the perioperative period.¹ Specifically, intraoperative arterial hypotension (IOH), often defined as a MAP of less than 65 mm Hg, occurs in more than 75% of patients undergoing surgery.² Several factors, including induction of general anaesthesia, positive pressure ventilation, central sympathectomy, surgical manipulation, and perioperative bleeding contribute to IOH.³ Retrospective data report an association between IOH and adverse postoperative outcomes, including major adverse cardiovascular events (MACE) and death.^{3,4} Furthermore, increased odds of organ damage of less severity, such as acute kidney injury (AKI) and myocardial injury, have been observed with increased cumulative duration of IOH below a MAP of 65 mm Hg.⁴ Although the literature consistently reports an association between IOH and adverse postoperative outcomes, most of the evidence relies on observational data.³ Residual and sometimes unknown confounding may limit the accuracy and generalisability of these results, for example, increased surgical duration is often associated with both the occurrence of IOH and the risk of MACE.⁵ RCTs can mitigate this confounding, although methodological challenges may impact the robustness of the results. For example, although researchers aim to generate group contrast between higher- and lower-target MAP groups in trials, sufficient group separation in achieved MAP may not always occur.⁶ This overlap between study groups may minimise observed effect sizes, preventing the assessment of a true dose-response or causal relationship between intraoperative blood pressure management and its subsequent impact on clinical outcomes.³ Consequently, summarising the impact of such group separation on cardiovascular outcomes is warranted.

A previous meta-analysis summarising evidence on the association between IOH and postoperative outcomes considered a limited number of RCTs in the noncardiac surgical population ($n=4$),⁶ did not evaluate group separation, and did not include recently published methodologically rigorous trials with improved intervention adherence and outcome reporting.^{7–9} Therefore, we aimed to provide an up-to-date, comprehensive synthesis of randomised evidence, to better elucidate the relationship between IOH and postoperative mortality and cardiovascular outcomes in patients undergoing noncardiac surgery.

This systematic review and meta-analysis aimed to summarise and analyse evidence from RCTs that categorise groups of patients based on MAP thresholds, examining the impact of IOH on all-cause mortality, AKI, and myocardial injury in patients undergoing noncardiac surgery. Furthermore, we aimed to assess achieved MAP differences and differences in exposure below MAP thresholds between intervention (i.e. 'higher-pressure' target) and control (i.e. 'lower-pressure' target) groups to evaluate achieved group contrast across the RCTs.

Methods

The systematic review was registered in PROSPERO (CRD420251230069) and is reported according to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines.¹⁰

Outcomes

The primary outcome of this meta-analysis was all-cause mortality within 30 days of surgery. Secondary outcomes included AKI and myocardial injury within 30 days of surgery. Given the potential lack of separation between MAP groups, the mean or median MAP achieved in the intervention and control arms of the included trials was analysed as a secondary outcome. When reported in the original RCT, depth and duration of hypotension, the minimum and maximum observed MAP, percentage of time observed under a pressure threshold, time-weighted average under MAP thresholds, and area under the MAP threshold (i.e. AUC) between intervention and control groups were also reported.

Eligibility criteria

Eligibility criteria were: RCTs that categorised groups based on MAP thresholds for adult patients undergoing noncardiac surgery (excluding patients <18 yr age) and reporting on at least one of the following outcomes: all-cause mortality, AKI, or myocardial injury.

Search strategy

Three investigators (MS, JA, DS) independently screened titles and abstracts using a standardised form to assess eligibility, followed by full-text review to confirm inclusion. Discrepancies were resolved by discussion with two investigators (MS, WVK). Databases included PubMed, Embase, and Web of Science. Search strings combined keywords and Medical Subject Headings related to IOH, blood pressure management, and noncardiac surgery. This search included studies up to November 2025. The search strategy can be found in [Box 1](#).

Box 1**Search strategy.**

("Hypotension"[Mesh] OR hypotens* OR "blood pressure drop" OR "blood pressure decrease*" OR "low blood pressure" OR "mean arterial pressure" OR MAP OR "arterial hypotension") AND ("Intraoperative Period"[Mesh] OR intraoperativ* OR intra-operativ* OR perioperativ* OR intra-anaesthetic OR anesthe* OR anaesthetic* OR surgery OR surgical OR operation* OR operative) AND ("Acute Kidney Injury"[Mesh] OR "Myocardial Injury"[Mesh] OR "Myocardial Infarction"[Mesh] OR stroke OR cerebrovascular accident* OR "organ injur*" OR "major adverse events" OR mortality OR death)

Data extraction and intervention characterisation

Three investigators (MS, JA, DS) independently extracted data using a standardised form. Extracted data included study characteristics, patient characteristics, type of surgery, sample size, intervention (i.e. 'higher-pressure') and control (i.e. 'lower-pressure') MAP targets, duration and depth of hypotension, and reported postoperative outcomes. Data on intraoperative haemodynamic management, including vasopressor and fluid administration, were recorded when available. Studies included in the final review and analysis consisted of either a fixed or personalised strategy to manage intraoperative blood pressure. For each included study, two groups were identified: (1) an intervention ('higher-pressure' target) group, and (2) a control ('lower-pressure' target) group. For example, trials that compared hypertension vs hypotension avoidance were classified as follows: hypertension avoidance strategies were categorised as the control ('lower-pressure' target) group, while hypotension avoidance strategies were categorised as the intervention ('higher-pressure' target) group.

Data quality assessment

Similar to Wesselink and colleagues,³ an adapted version of the Consolidated Standard of Reporting Trials (CONSORT) was used to assess data quality. This data quality tool was used to assess methodological characteristics in the following categories: internal validity, external validity, bias, and precision. Each criterion was assigned to a category or categories scored as + (positive), - (negative), or ? (unclear). The revised Cochrane risk of bias tool for RCTs was used by three investigators (MS, JA, DS) to independently assess the risk of bias for each included trial.¹¹ Overall quality of the evidence was assessed using the Grading of Recommendations Assessment, Development, and Evaluation (GRADE).¹²

Statistical analysis

Meta-analyses were conducted in R (version 4.2.3; Vienna, Austria) using the 'metafor' package. For the analysis of the primary and secondary outcomes, study-level log risk ratios [log(RR)] and their variances were calculated for each RCT. Point

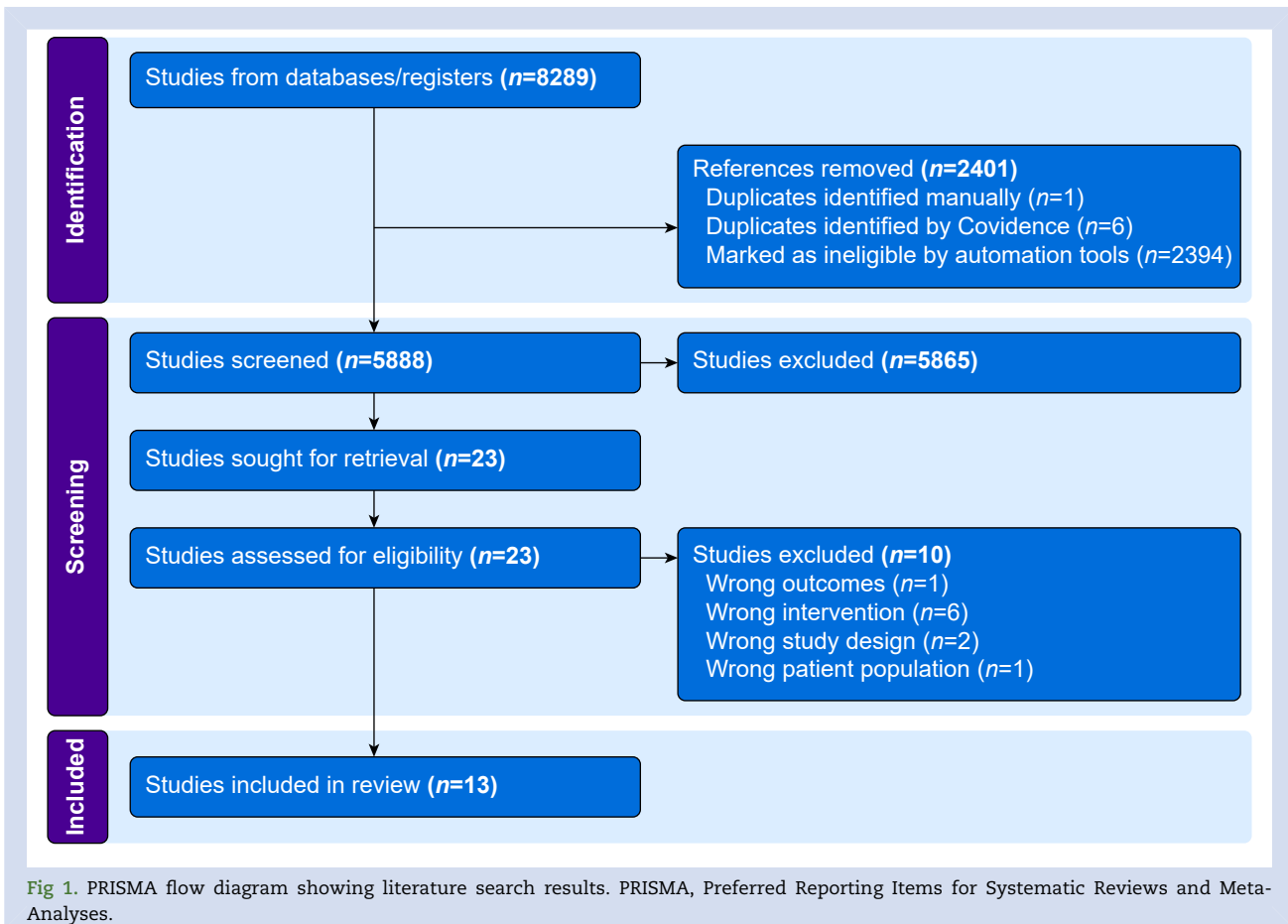
estimates for binary outcomes were expressed as an odds ratio, risk ratio, or hazard ratio, with 95% confidence intervals using the Mantel-Haenszel method.¹³ Continuous outcomes were expressed as a mean difference with 95% confidence intervals using the inverse variance method.¹⁴ If the measures of central tendency and variability were only expressed as the median and interquartile range, Wan's method was used to estimate both the mean and standard deviation.¹⁵ A two-sided Cochran's Q test, with $\alpha=0.1$ was used to test the heterogeneity hypothesis. Higgins and Thompson's I^2 was used to quantify statistical consistency. Values of I^2 were interpreted as follows: low (<25%), moderate (25–75%), and high (>75%), with respect to statistical consistency.¹⁶ A threshold of $I^2 > 50\%$ was used to signify statistical heterogeneity. A fixed-effect models was pursued with $I^2 < 50\%$, while $I^2 > 50\%$ supported use of a random-effects model. A $P < 0.05$ was considered statistically significant. The results of the highest vs lowest MAP groups were extracted from the studies if multiple MAP threshold groups were investigated in the original RCT.

Exploratory analysis used a meta-regression to examine whether the difference in achieved MAP (difference in mean or median, as reported in Table 2) between intervention (i.e. 'higher-pressure') and control ('lower-pressure') groups modified the primary and secondary outcomes. A separate exploratory meta-regression model used the difference in mean or median AUC <65 mm Hg (as reported in Table 3) as an effect modifier of the study outcomes.

A trial sequential analysis was performed to evaluate the clinical relevance of the pooled effect estimate for the meta-analysis of the primary and secondary outcomes.^{17,18} Required information sizes (RIS) considered three scenarios: 1) a relative risk reduction (RRR) of 5%, 2) a RRR of 10%, and 3) a RRR of 20%. The 5% RRR corresponds to an absolute risk reduction of ~0.15% (number needed to treat (NNT)~670). The 10% RRR corresponds to an absolute risk reduction of ~0.3% (NNT~330), while the 20% RRR corresponds to an absolute risk reduction of ~0.6% (NNT~200). These three scenarios reflect small-to-moderate absolute effects for clinically meaningful outcomes (e.g. mortality, stroke). A cumulative Z-statistic was calculated sequentially and plotted with the cumulative information size.^{17,18} O'Brien-Fleming monitoring boundaries were calculated as a function of the cumulative information fraction to assess whether the evidence included in this study crossed the boundaries of benefit or harm. Statistical significance was assessed based on whether the cumulative Z-curve crossed monitoring boundaries before reaching the RIS. This trial sequential analysis used $\alpha=0.05$, with 80% power ($\beta=0.20$).

Results**Study characteristics**

The electronic database search identified nine RCTs, published from 2017 to 2025, that met the inclusion criteria for the primary outcome ($n=14\,658$) (Fig 1, Table 1), with 13 RCTs included in analyses of the secondary outcomes.^{7–9,19–25} Eight RCTs were completed in Europe,^{7,8,19–21,24–26} three in Asia,^{9,23,27} and two were multicontinental.^{22,28} Included participants underwent major noncardiac surgery, with specific surgical procedures including abdominal, orthopaedic, and other elective procedures. General anaesthesia was the most common form of anaesthetic management, often induced with propofol. Vasoactive



medications were used to maintain target blood pressure, including norepinephrine, phenylephrine, and ephedrine.

Effect of higher vs lower mean arterial pressure targets on mortality

The primary outcome occurred in 78 of 7327 participants randomly assigned to intervention ('higher-pressure' target) study groups, compared with 75 of 7331 participants assigned to control ('lower-pressure' target) study groups (Table 2). The fixed-effect model showed no significant difference between intervention and control groups in the primary outcome across the nine RCTs that reported mortality within 30 days of surgery (Fig 2). The pooled log risk ratio was 0.06 (95% CI: -0.6 to 0.37, $P=0.73$), corresponding to a risk ratio of 1.06 (95% CI: 0.77 to 1.45). Between-study heterogeneity was negligible ($I^2=0\%$, $Q=4.77$, $P=0.78$), indicating that effect sizes were consistent across the nine studies.

Trial sequential analysis for an RRR of 5% revealed RIS of 1 192 582, an RRR of 10% yielded RIS of 290 569 participants, while RIS of 68 848 was calculated for an RRR of 20%. The cumulative Z-curve did not cross O'Brien-Fleming monitoring boundaries in all scenarios (Supplementary Fig 5). These results demonstrate that, based on the available data, there is insufficient evidence to conclude there is no effect if a 20% RRR or lower is considered clinically relevant, as the current cumulative sample size is well below the RIS for these scenarios.

Effect of higher vs lower mean arterial pressure targets on acute kidney injury and myocardial injury

AKI occurred in 866 of 7191 participants randomly assigned to intervention study groups, compared with 861 of 7181 participants assigned to control study groups (RR: 1.00, 95% CI: 0.92 to 1.09) (Supplementary Fig 1). Stratifying the results by differences in achieved MAP did not modify the effect on the outcome of AKI (Supplementary Fig 3).

Myocardial injury occurred in 845 of 7260 participants randomly assigned to intervention study groups, compared with 823 of 7268 participants assigned to control study groups (RR: 1.03, 95% CI: 0.94 to 1.12) (Supplementary Fig 2). Stratifying the results by differences in achieved MAP did not modify the effect on the outcome of myocardial injury (Supplementary Fig 4).

The cumulative Z-curve did not cross O'Brien-Fleming monitoring boundaries for RRRs of 5% and 10% for both AKI and myocardial injury. However, RIS boundaries were crossed for an RRR of 20% (Supplementary Figs. 6 and 7).

Differences in achieved mean arterial pressure

Hypotensive burden varied across the included trials (Tables 2 and 3). Nicklas and colleagues²⁴ reported one of the largest differences in hypotensive exposure, while smaller differences in group contrast were reported by Zhao and colleagues⁹. Evidence from Saugel and colleagues⁷ and Kant and colleagues⁸ shows differences in the area under MAP

Table 1 Characteristics of the 13 studies included in the meta-analysis. ACE, angiotensin converting enzyme; ARB, angiotensin-II receptor blocker; BIS, bispectral index; ENT, ear-nose-throat; GA, general anaesthesia; SBP, systolic blood pressure; TCI, target controlled infusion.

Study	Country	Anaesthesia induction	Anaesthesia maintenance	Blood pressure management—intervention	Blood pressure management—control	Type of surgery	Target pressure
Futier and colleagues 2017 ¹⁹	France	General anaesthesia	Balanced anaesthesia, protocolised fluid therapy, norepinephrine	Norepinephrine	Ephedrine	Major abdominal	Intervention: SBP within + or –10% of patient's resting SBP Control: SBP $\geq$ 80 mm Hg or $\geq$ 60% of baseline
Langer and colleagues 2019 ²⁰	Italy	General anaesthesia (propofol, opioids)	Balanced anaesthesia with volatile agents or propofol; clinician-directed fluids/vasopressors	Fluids, vasopressors at clinician discretion	Routine care without predefined target with fluids, vasopressors at clinician discretion	General, urological, thoracic, orthopaedic	Intervention: MAP $\geq$ 90% of baseline MAP Control: No target
Wanner and colleagues 2021 ²¹	Switzerland	General anaesthesia (propofol TCI or volatile agents, opioids (fentanyl, remifentanyl)	Balanced anaesthesia, BIS-guided depth, epidural	Ephedrine, norepinephrine	Managed per institutional policy	Major noncardiac (vascular, thoracic, urologic, visceral, orthopaedic)	Intervention: MAP $\geq$ 75 mm Hg Control: MAP $\geq$ 60 mm Hg
Marcucci and colleagues 2023 ²² and Garg and colleagues 2025 ²⁸	Multinational (110 hospitals, 22 countries)	General anaesthesia	Volatile or IV agents, BIS monitoring not specified	Fluids, vasopressors	Standard clinical practice (not detailed)	Major noncardiac (general, vascular, orthopaedic, thoracic, urologic, gynaecological)	Intervention: MAP $\geq$ 80 mm Hg Control: MAP $\geq$ 60 mm Hg
Nicklas and colleagues 2024 ²⁴	Germany	General anaesthesia	General anaesthesia with routine monitoring	Fluids, vasopressors	Fluids, vasopressors	Major noncardiac (abdominal, orthopaedic, vascular)	Intervention: MAP $\geq$ individual preoperative baseline MAP Control: MAP $\geq$ 65 mm Hg
Bergholz and colleagues ²⁶	Germany	General anaesthesia	General anaesthesia	Fluids, vasopressors	Fluids, vasopressors	Major noncardiac (abdominal, orthopaedic)	Intervention: MAP near individual baseline Control: MAP $\geq$ 65 mm Hg
Pang and colleagues 2025 ²³	China	General anaesthesia	General anaesthesia	Norepinephrine, fluids	Fluids, vasopressors	Major noncardiac (gastrointestinal)	Intervention: MAP or SBP within + or – 10–20% of baseline Control: MAP $\geq$ 65 mm Hg
Luney and colleagues 2025 ²⁵	UK	General anaesthesia (propofol or etomidate), spinal anaesthesia (bupivacaine)	Volatile agents (sevoflurane), GA maintenance, regional blocks (femoral or fascia iliaca)	Ephedrine, metaraminol, IV fluids	Standard clinical practice (not detailed)	Hip fracture repair	Intervention: SBP $\geq$ 80% baseline and MAP $\geq$ 75 mm Hg Control: No target (usual care)
Zhao, and colleagues 2025 ⁹	China	General anaesthesia with propofol	Propofol or sevoflurane	Norepinephrine infusion	Standard clinical practice (not detailed)	Major abdominal (oesophageal, gastric, hepatopancreatobiliary, intestinal, colorectal, gynaecological, urologic)	Intervention: $\geq$ 80 mm Hg Control: $\geq$ 65 mm Hg or $\geq$ 60% baseline MAP
Kant and colleagues 2025 ⁸	Netherlands	General anaesthesia, spinal	General anaesthesia, epidural	Vasopressors (discretion of the anaesthesiologist, commonly used phenylephrine)	Managed per institutional policy, e.g. ACE inhibitors or ARBs withheld day of surgery	Elective noncardiac (general, gynaecological, orthopaedic, ENT, neurosurgery, urologic, plastic)	Intervention: - Low risk: MAP $\geq$ 70 mm Hg - Intermediate Risk: MAP $\geq$ 80 mm Hg - High Risk: MAP $\geq$ 90 mm Hg Control: $\geq$ 65 mm Hg

Continued

Table 1 Continued

Study	Country	Anaesthesia induction	Anaesthesia maintenance	Blood pressure management—intervention	Blood pressure management—control	Type of surgery	Target pressure
Saugel and colleagues 2025 ⁷	Germany	General anaesthesia	Balanced or total i.v. anaesthesia; epidural	Norepinephrine	Standard clinical practice (not detailed)	Major abdominal (general, urologic, gynaecological, liver)	Intervention: ≥Preoperative mean night time MAP Control: ≥65 mm Hg Intervention: MAP within + or −20% of baseline Control: MAP ≥ 65 mm Hg
Fang and colleagues 2025 ²⁷	China	General anaesthesia	General anaesthesia	Norepinephrine	Vasopressors	Major noncardiac (abdominal)	

Table 2 Primary outcome (all-cause 30-day mortality) and effect sizes for the nine studies included in the meta-analysis. *Italicised terms indicate the average achieved pressure was reported. Bold terms indicate the median achieved pressure was reported.*

Study	Event number in Intervention ('high- pressure' target) group	Sample size of Intervention ('high' pressure target) group	# Of events in Control ('low' pressure target) group	Sample size of Control ('low' pressure target) group	Achieved pressure of Intervention ('high' pressure target) group	Achieved pressure of control ('low' pressure target) group	Risk ratio (95% confidence interval)
Futier and colleagues 2017 ¹⁹	9	147	8	145	81 mm Hg	75 mm Hg	1.11, 0.43–2.80
Wanner and colleagues 2021 ²¹	1	225	1	226	81 mm Hg	77 mm Hg	1.00, 0.06–15.96
Marcucci and colleagues 2023 ²²	50	3742	43	3748	86 mm Hg	81 mm Hg	1.16, 0.78–1.75
Nicklas and colleagues 2024 ²⁴	0	166	3	158	92 mm Hg	72 mm Hg	0.14, 0.01–2.61
Pang and colleagues 2025 ²³	1	89	0	90	86 mm Hg	72 mm Hg	3.03, 0.13–73.48
Luney and colleagues 2025 ²⁵	1	34	0	30	94 mm Hg	77 mm Hg	2.66, 0.11–62.87
Zhao, and colleagues 2025 ⁹	6	739	5	738	87 mm Hg	82 mm Hg	1.20, 0.37–3.91
Kant and colleagues 2025 ⁸	5	1618	5	1629	87 mm Hg	81 mm Hg	1.01, 0.29–3.47
Saugel and colleagues 2025 ⁷	5	567	10	567	89 mm Hg	80 mm Hg	0.50, 0.17–1.45

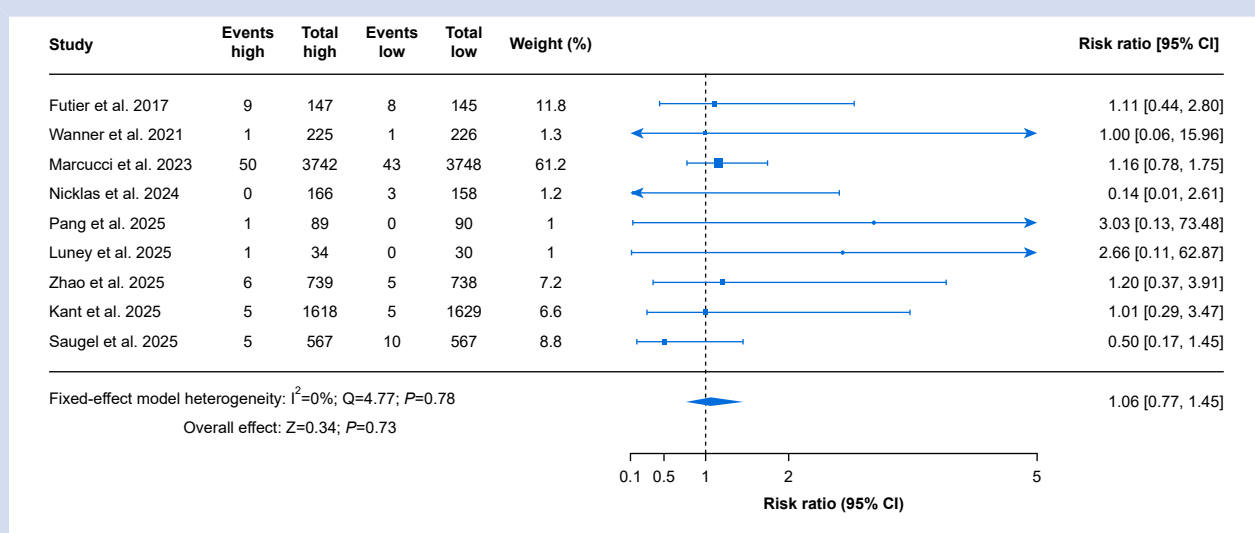


Fig 2. Forest plot summarising the primary outcome (all-cause mortality) across nine trials using a fixed-effects model for higher vs lower blood pressure targets. The pooled estimate shows no significant effect (RR: 1.06, 95% CI: 0.77–1.45, $\log(RR)=-0.06$, 95% CI: -0.26 to 0.37 , $z=0.34$), with negligible heterogeneity ($I^2=0\%$). The size of each point estimate square reflects the proportional impact of the individual study's weight on the meta-analysis.

thresholds of 60–70 mm Hg between intervention and control groups. However, none of these differences in exposure to hypotension was associated with primary and secondary outcomes. For example, Fig 3 demonstrates no effect modification by differences in achieved MAP on mortality (estimate -0.09 per mm Hg, 95% CI -0.23 to 0.05 , $P=0.86$). Six studies (Table 3) reported AUC between intervention and control groups, consistently using an AUC threshold of <65 mm Hg.^{7–9,23,24,26} Greater hypotension burden in the control compared with the intervention group was not associated with the study outcomes.

Quality assessment

Adequate reporting according to the adapted version of CONSORT from Wesselink et al.,³ was demonstrated by Saugel and colleagues,⁷ Kant and colleagues,⁸ Zhao et al., Marcucci and colleagues,²² and Futier and colleagues.¹⁹ A pilot study by Luney and colleagues,²⁵ demonstrated adequacy, except for its limited sample size, and incomplete reporting of allocation concealment. Reporting on harms and allocation concealment were the most frequently underreported domains. Overall, study quality was assessed as moderate to high, with none of the RCTs demonstrating a risk of bias warranting exclusion from the analysis. Low certainty evidence primarily stemmed from imprecision (e.g. wide confidence intervals) and the lack of blinding of study personnel (Supplementary Tables 1–3).

Discussion

In this systematic review and meta-analysis of RCTs, targeting higher vs lower intraoperative MAP did not lead to a significant difference in the primary outcome of all-cause 30-day mortality, and secondary outcomes of AKI, and myocardial injury. The negligible between-study heterogeneity indicates consistent findings across the study populations and surgical settings.

Including only RCTs minimised residual confounding, which previously limited reviews including observational studies that often linked IOH with adverse postoperative outcomes.³ The results are consistent with previous evidence from D'Amico and colleagues,⁶ who also found no meaningful reduction in mortality or MACE with permissive (MAP ≤ 60 mm Hg) vs targeted (MAP >60 mm Hg) MAP in a meta-analysis in both cardiac and noncardiac surgery patients. However, the inclusion of several recently published adequately sized RCTs in noncardiac surgery in the present analysis provides more robust estimates.²⁹

Three major differences exist between the methodology of this meta-analysis and that of D'Amico and colleagues.⁶ Firstly, this meta-analysis only included elective noncardiac surgeries, whereas study populations from D'Amico and colleagues were heterogeneous. For example, D'Amico and colleagues included a trial of haemorrhage control in patients presenting with penetrating trauma. The inclusion of these high-acuity surgical presentations may result in differences related to physiology and management compared with the other included trials in D'Amico and colleagues which included non-urgent cases. Another difference exists with the inclusion of study outcomes in D'Amico and colleagues, which ranged from delirium to mortality—outcomes that clearly differ in pathophysiology and clinical relevance. The inclusion of heterogeneous study outcomes from D'Amico and colleagues reflected the limited number of studies at the time in the noncardiac surgical population. Recent evidence, for example from Kant and colleagues⁸ and Saugel and colleagues,⁷ with consistent reporting of robust study outcomes, is reflected in the current meta-analysis. Lastly, this meta-analysis included additional findings related to group separation between study targets, which identify areas for improvement for future trials in the reporting of and adherence to haemodynamic separation. Additionally, the reporting on haemodynamic separation in this meta-analysis identifies that the lack of a statistically significant effect between 'higher' compared with 'lower' intraoperative pressure

Table 3 Secondary haemodynamic outcomes and effect sizes for the included studies. *Note: Haemodynamic measures for Marcucci and colleagues 2023²² also reflect the haemodynamic measures of Garg and colleagues 2025²⁸ which is a substudy of Marcucci and colleagues 2023²²

Study	Secondary Haemodynamic Measure	Intervention ('higher-pressure' target) group	Control ('lower-pressure' target) group
Futier and colleagues 2017 ¹⁹	Not reported	-	-
Langer and colleagues 2019 ²⁰	Proportion of time spent $\geq 90\%$ of baseline MAP	Mean 65% (SD 25%)	Mean 29% (SD 28%)
Wanner and colleagues 2021 ²¹	Time spent <75 mm Hg	Median 72 min (IQR 41–238 min)	Median 121 min (IQR 70–210 min)
Wanner and colleagues 2021 ²¹	Time spent <65 mm Hg	Median 9 min (IQR 3–24 min)	Median 23 min (IQR 8–49 min)
Wanner and colleagues 2021 ²¹	Minimal MAP of hypotensive episodes	Median 60 mm Hg (IQR 55–63 mm Hg)	Median 57 mm Hg (IQR 51–62 mm Hg)
Wanner and colleagues 2021 ²¹	Duration of hypotensive episodes	Median 80 sec (IQR 53–145 sec)	Median 116 sec (IQR 81–240 sec)
Marcucci and colleagues 2023 ^{22*}	Cumulative time MAP <60 mm Hg	Mean 2.8 min (SD 0.3 min)	Mean 4.7 min (SD 2.6 min)
Marcucci and colleagues 2023 ^{22*}	Mean difference in achieved MAP	–1.96 min (95% CI: –2.49 to 1.44)	-
Nicklas and colleagues 2024 ²⁴	AUC MAP <65 mm Hg	Median 0 mm Hg min (IQR 0–35)	Median 35 mm Hg min (IQR 0–138)
Nicklas and colleagues 2024 ²⁴	Proportion of time $\geq$ baseline MAP	Median 57% (IQR 35–73%)	Median 7% (IQR 2–24%)
Luney and colleagues 2025 ²⁵	Time spent MAP <75 mm Hg	Mean 12.5 min h ⁻¹ of surgery	Mean 18.8 min h ⁻¹ of surgery
Luney and colleagues 2025 ²⁵	Time spent SBP <80% baseline	Mean 13 min h ⁻¹ of surgery	Mean 24 min h ⁻¹ of surgery
Zhao, and colleagues 2025 ⁹	Cumulative MAP <65 mm Hg	Median 1 min (IQR 0–7 min)	Median 8 min (IQR 0–20)
Zhao, and colleagues 2025 ⁹	Minimal achieved MAP	Median 68 mm Hg (IQR 63–73 mm Hg)	Median 63 mm Hg (IQR 59–67 mm Hg)
Zhao, and colleagues 2025 ⁹	AUC MAP <70 mm Hg	Median 23 mm Hg min (IQR 0–96 mm Hg min)	Median 125 mm Hg min (IQR 30–276 mm Hg min)
Zhao, and colleagues 2025 ⁹	AUC MAP <65 mm Hg	Median 3 mm Hg min (IQR 0–25 mm Hg min)	Median 23 mm Hg min (IQR 0–87 mm Hg min)
Zhao, and colleagues 2025 ⁹	AUC MAP <60 mm Hg	Median 0 mm Hg min (IQR 0–4 mm Hg min)	Median 1 mm Hg min (IQR 0–20 mm Hg min)
Zhao, and colleagues 2025 ⁹	AUC MAP >110 mm Hg	Median 32 mm Hg min (IQR 0–128 mm Hg min)	Median 15 mm Hg min (IQR 0–83 mm Hg min)
Kant and colleagues 2025 ⁸	Proportion of time <65 mm Hg	Mean 47.7% (SD 30.7%)	Mean 71% (SD 31.3%)
Kant and colleagues 2025 ⁸	AUC MAP <65 mm Hg	Mean 28.5 mm Hg min (SD 66 mm Hg min)	Mean 70.6 mm Hg min (SD 112.1 mm Hg min)
Kant and colleagues 2025 ⁸	Proportion of time MAP <60 mm Hg	Mean 19.7% (SD 25.8%)	Mean 39.7% (SD 36.2%)
Kant and colleagues 2025 ⁸	AUC MAP <60 mm Hg	Mean 6 mm Hg min (SD 18.3 mm Hg min)	Mean 30.7 mm Hg min (SD 65.4 mm Hg min)
Kant and colleagues 2025 ⁸	Proportion of time MAP <70 mm Hg	Mean 66.5% (SD 31%)	Mean 84.4% (SD 25.4%)
Kant and colleagues 2025 ⁸	AUC MAP <70 mm Hg	Mean 70.7 mm Hg min (SD 112.4 mm Hg min)	Mean 142.6 mm Hg min (SD 169.2 mm Hg min)
Kant and colleagues 2025 ⁸	Time-weighted average MAP	Mean 77 mm Hg (SD 9.9 mm Hg)	Mean 72.3 mm Hg (SD 9.4 mm Hg)
Saugel and colleagues 2025 ⁷	AUC MAP <65 mm Hg	Median 6 mm Hg min (IQR 0–54 mm Hg min)	Median 48 mm Hg min (IQR 7–151 mm Hg min)
Saugel and colleagues 2025 ⁷	Proportion of time $\geq$ preoperative mean night-time MAP	Median 79% (IQR 62–91%)	Median 57% (IQR 28–81%)
Pang and colleagues 2025 ²³	AUC MAP <65 mm Hg	Median 0 mm Hg min (IQR 0–1.88)	Median 41.2 mm Hg min (IQR 16.5–80.6)
Pang and colleagues 2025 ²³	Cumulative time MAP <75 mm Hg	Median 11.5 min (IQR 3.1–28.2)	Median 112 min (IQR 81–140)
Pang and colleagues 2025 ²³	Cumulative time MAP <65 mm Hg	Median 0 min (IQR 0 – –0.76)	Median 16.7 min (IQR 8.0–26.3)
Pang and colleagues 2025 ²³	Cumulative time MAP <60 mm Hg	Median 0 min (IQR 0–0)	Median 2.08 min (IQR 0.63–5.67)
Bergholz and colleagues 2025 ²⁶	AUC MAP <65 mm Hg	Median 20 mm Hg min (IQR 4–54)	Median 143 mm Hg min (IQR 34–191)
Bergholz and colleagues 2025 ²⁶	Proportion of time $\geq$ individualised MAP target	Median 59% (IQR 41–81%)	Not reported
Bergholz and colleagues 2025 ²⁶	Time-weighted average MAP below target	Median 3.29 mm Hg (IQR: 1.64–6.82)	Not reported

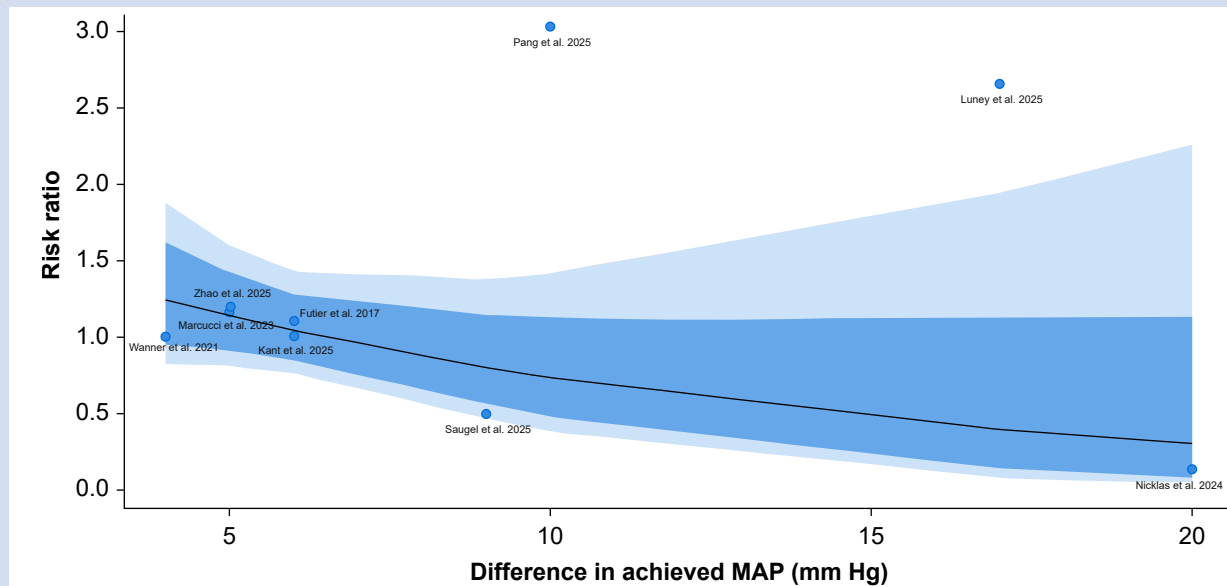


Fig 3. Relationship between the difference in achieved mean arterial pressure (MAP) between intervention (i.e. 'higher-pressure' target) and control (i.e. 'lower-pressure' target) groups and the risk ratio of the primary outcome. Each point represents an individual randomised trial, with point size scaled to study precision. The black line shows the predicted estimate from the fixed-effect meta-regression model, with the shaded regions representing the 80% and 95% confidence intervals. There was no significant association between the magnitude of the achieved MAP difference and the risk ratio for the primary outcome (estimate = -0.09 per mm Hg, 95% CI -0.23 to 0.05 , $P=0.86$).

groups may be because of limited group separation. The results of this meta-analysis expand on previous evidence regarding group separation, by providing an exploratory analysis linking differences in achieved MAP, and AUC of 65 mm Hg, with the outcomes of all-cause mortality, AKI, and myocardial injury. Despite the inclusion of more evidence compared with D'Amico and colleagues,⁶ trial sequential analysis was unable to identify a clinically relevant effect for the primary outcome considering RRRs of 20% or less. For secondary outcomes of acute kidney and myocardial injury, RIS boundaries were crossed considering a RRR of 20%. Still, the results should be interpreted with caution, as they may be limited by the lack of sufficient MAP separation between groups.

Although statistically significant differences were observed in the mean or median achieved MAP between study groups in some of the included trials, the magnitude of these differences seems limited (~ 7 – 15 mm Hg). Differences in achieved MAP or AUC of 65 mm Hg displayed in the meta-regressions between intervention and control groups were not associated with the study outcomes. The separation in achieved average MAP was modest, and in most studies less than 10 mm Hg. Often, the included RCTs identified a control group as usual care, where clinicians often prevent MAP from falling below 65 mm Hg, which may have limited the ability to detect clinically relevant effects.^{3,6} Although a 10 mm Hg difference in achieved MAP (i.e. the average across the included trials) between study groups may be considered meaningful group separation, the observational literature suggests that a MAP below 60–65 mm Hg for extended cumulative durations is associated with myocardial injury and AKI.^{3,4} However, the minimum achieved pressures across the included RCTs did not frequently fall below this threshold. Therefore, achieved group separation

consequently limits the ability of randomised evidence to demonstrate potential causality between IOH (MAP below 60–65 mm Hg) and adverse postoperative outcomes.

This lack of meaningful group separation between 'higher-pressure' and 'lower-pressure' groups may be explained by years of observational evidence linking IOH to adverse outcomes.^{3,4,30} The observational non-causal relationship has likely shaped clinical practice, emphasising the importance of adequate pressures as a surrogate of adequate organ perfusion, and that intentionally increasing exposure to hypotension is unethical.

The absence of benefit with higher MAP targets raises questions about the causal role of the magnitude of blood pressure in perioperative organ injury. While low blood pressure has been associated with adverse outcomes, it may serve as a signal of inherent patient risk to postoperative outcomes rather than a direct causal relationship.^{31,32} From a physiological perspective, once a minimum perfusion pressure is achieved, increases in arterial pressure may not meaningfully impact organ perfusion or function. Furthermore, interventions used to maintain higher pressures, such as vasopressors or excessive fluid administration, may induce competing risks, including myocardial strain, arrhythmias, and fluid overload, potentially negating any advantage of improved organ perfusion.^{33,34}

Therefore, the results of this review do not necessarily suggest that IOH is not associated with adverse postoperative outcomes; instead, the results potentially demonstrate that higher pressure targets compared with usual care do not lead to differences in adverse postoperative outcomes, including in patients considered at increased risk.⁸ However, the overlap in achieved pressures between intervention and control groups prevents firm conclusions.

Limitations

Most trials reported the depth and duration of IOH; however, this information was inconsistently defined or incompletely reported in some instances, limiting the ability to determine if the degree of hypotension was associated with all-cause mortality, AKI, or myocardial injury. For example, the mean or median achieved pressures were reported in all included studies; however, only the amount or proportion of time under MAP thresholds or AUC were reported in some instances. The relatively small number of included studies also limits the quality of evidence surrounding the relationship between IOH and the study outcomes.

The lack of MAP separation between intervention and control groups can be attributed to the design of most trials, which used usual care as the control, which often constitutes the maintenance of MAP above 65 mm Hg.^{20,25} Similarly, some included studies also listed the 'control' group as hypertension avoidance, establishing the criterion for this group as MAP >60 mm Hg, leading to overlap with the intervention arms for hypotension avoidance, which were often established as MAP >80 mm Hg.^{8,9,21,22}

Conclusions

The neutral findings of this review should not be interpreted as evidence that intraoperative haemodynamic management is an unimportant intervention. Instead, the findings of this review reflect that increasing intraoperative blood pressure above that targeted in usual care, were not associated with reductions in adverse postoperative outcomes, limiting the ability to identify a dose-response or causal relationship. At minimum, IOH acts as an indicator of underlying physiological stress; however, strategies that simply raise intraoperative arterial pressure, but do not achieve adequate MAP separation between study groups, will likely attenuate the impact of higher vs lower intraoperative arterial pressures for postoperative outcomes. Future studies should be designed to achieve and sustain greater differences in intraoperative MAP, and the strategies required to achieve group separation in trials investigating intraoperative blood pressure management require further investigation. Additionally, rigorous reporting that highlights group separation in MAP is rarely reported across the included RCTs, limiting the ability to contrast differences in achieved MAP and to identify if hypotensive burden is associated with adverse outcomes.

Authors' contributions

First draft of the manuscript: MS

Systematic search: MS

Study selection: MS, DS, JA

Data extraction: MS, DS, JA

Statistical analysis: MS

Design and focus of the manuscript: MS, WVK, JS, DPV, TK, MK

Feedback on the manuscript: SAM, IR, DSM, and the remaining co-authors

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Covidence automation tools to assist with prioritisation and identification of irrelevant records; all inclusion and exclusion

decisions were confirmed by human reviewers. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of interest

The authors declare that they have no conflicts of interest.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.bja.2026.05.061>.

References

1. Südfeld S, Brechnitz S, Wagner JY, et al. Post-induction hypotension and early intraoperative hypotension associated with general anaesthesia. *Br J Anaesth* 2017; **119**: 57–64
2. Bijker JB, Van Klei WA, Kappen TH, Van Wolfswinkel L, Moons KGM, Kalkman CJ. Incidence of intraoperative hypotension as a function of the chosen definition: literature definitions applied to a retrospective cohort using automated data collection. *Anesthesiology* 2007; **107**: 213–20
3. Wesselink EM, Kappen TH, Torn HM, Slooter AJC, Van Klei WA. Intraoperative hypotension and the risk of postoperative adverse outcomes: a systematic review. *Br J Anaesth* 2018; **121**: 706–21
4. Salmasi V, Maheshwari K, Yang D, et al. Relationship between intraoperative hypotension, defined by either reduction from baseline or absolute thresholds, and acute kidney and myocardial injury after noncardiac surgery: a retrospective cohort analysis. *Anesthesiology* 2017; **126**: 47–65
5. Gregory A, Stapelfeldt WH, Khanna AK, et al. Intraoperative hypotension is associated with adverse clinical outcomes after noncardiac surgery. *Anesth Analg* 2021; **132**: 1654–65
6. D'Amico F, Fominskiy EV, Turi S, et al. Intraoperative hypotension and postoperative outcomes: a meta-analysis of randomised trials. *Br J Anaesth* 2023; **131**: 823–31
7. Saugel B, Meidert AS, Brunkhorst FM, et al. Individualized perioperative blood pressure management in patients undergoing major abdominal surgery: the IMPROVE-multi randomized clinical trial. *JAMA* 2025; **334**: 1893–904
8. Kant M, van Klei WA, Hollmann MW, et al. Proactive vs reactive treatment of hypotension during surgery: the PRE-TREAT randomized clinical trial. *JAMA* 2025; **334**: 1905–14
9. Zhao B, Zhang J, Xie Y, et al. Intensive vs conventional intraoperative blood pressure management on cardiovascular events after major abdominal surgery: the BP-CARES randomized trial. *J Am Coll Cardiol* 2025; **86**: 892–906
10. Page MJ, McKenzie JE, Bossuyt PM, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ* 2021; **372**: n71
11. Higgins JP, Green S. *Cochrane Handbook for Systematic Reviews of Interventions*. Chichester: John Wiley & Sons; 2008
12. Atkins D, Eccles M, Flottorp S, et al. Systems for grading the quality of evidence and the strength of recommendations I: critical appraisal of existing approaches The GRADE Working Group. *BMC Health Serv Res* 2004; **4**: 38
13. Moher D, Liberati A, Tetzlaff J, Altman DG. PRISMA Group. preferred reporting items for systematic reviews and meta-analyses: the PRISMA statement. *Int J Surg* 2010; **8**: 336–41

14. Davies HTO, Crombie IK, Tavakoli M. When can odds ratios mislead? *BMJ* 1998; **316**: 989–91
15. Wan X, Wang W, Liu J, Tong T. Estimating the sample mean and standard deviation from the sample size, median, range and/or interquartile range. *BMC Med Res Methodol* 2014; **14**: 135
16. Higgins JPT, Thompson SG, Deeks JJ, Altman DG. Measuring inconsistency in meta-analyses. *BMJ* 2003; **327**: 557–60
17. Wetterslev J, Thorlund K, Brok J, Gluud C. Trial sequential analysis may establish when firm evidence is reached in cumulative meta-analysis. *J Clin Epidemiol* 2008; **61**: 64–75
18. Wetterslev J, Jakobsen JC, Gluud C. Trial sequential analysis in systematic reviews with meta-analysis. *BMC Med Res Methodol* 2017; **17**: 39
19. Futier E, Lefrant J-Y, Guinot P-G, et al. Effect of individualized vs standard blood pressure management strategies on postoperative organ dysfunction among high-risk patients undergoing major surgery: a randomized clinical trial. *JAMA* 2017; **318**: 1346–57
20. Langer T, Santini A, Zadek F, et al. Intraoperative hypotension is not associated with postoperative cognitive dysfunction in elderly patients undergoing general anesthesia for surgery: results of a randomized controlled pilot trial. *J Clin Anesth* 2019; **52**: 111–8
21. Wanner PM, Wulff DU, Djurdjevic M, Korte W, Schnider TW, Filipovic M. Targeting higher intraoperative blood pressures does not reduce adverse cardiovascular events following noncardiac surgery. *J Am Coll Cardiol* 2021; **78**: 1753–64
22. Marcucci M, Painter TW, Conen D, et al. Hypotension-avoidance versus hypertension-avoidance strategies in noncardiac surgery: an international randomized controlled trial. *Ann Intern Med* 2023; **176**: 605–14
23. Pang Z, Liang S, Zhou N, et al. Individualized blood pressure regulation and acute kidney injury in older patients having major abdominal surgery: a pilot randomized trial. *Int J Surg* 2025; **111**: 2894–902
24. Nicklas JY, Bergholz A, Däke F, et al. Personalised blood pressure management during major noncardiac surgery and postoperative neurocognitive disorders: a randomised trial. *BJA Open* 2024; **11**, 100294
25. Luney MS, White SM, Moppett IK. Hip fracture intervention study for prevention of hypotension trial: a pilot randomized controlled trial. *A A Pract* 2025; **19**, e01891
26. Bergholz A, Grüber L, Khader WTAK, et al. Personalized perioperative blood pressure management in patients having major non-cardiac surgery: a bicentric pilot randomized trial. *J Clin Anesth* 2025; **100**, 111687
27. Fang Z, Wang Y-J, Zhu X-H, et al. Anti-inflammatory and anti-nociceptive effects of individualized blood pressure strategy based on low-dose noradrenaline infusion in elderly patients following major surgery: a randomized, controlled study. *J Clin Anesth* 2025; **107**, 111978
28. Garg AX, Marcucci M, Cuerden MS, POISE-3 Trial Investigators and Study Groups. A sub-study of the POISE-3 randomized trial examined effects of a perioperative hypotension-avoidance strategy versus a hypertension-avoidance strategy on the risk of acute kidney injury. *Kidney Int* 2025; **107**: 155–68
29. Powell M, Clark C, Alyakin A, Vogelstein JT, Hart B. Exploration of residual confounding in analyses of associations of metformin use and outcomes in adults with type 2 diabetes. *JAMA Netw Open* 2022; **5**, e2241505
30. Walsh M, Devereaux PJ, Garg AX, et al. Relationship between intraoperative mean arterial pressure and clinical outcomes after noncardiac surgery: toward an empirical definition of hypotension. *Anesthesiology* 2013; **119**: 507–15
31. Sessler DI, Bloomstone JA, Aronson S, et al. Perioperative quality initiative consensus statement on intraoperative blood pressure, risk and outcomes for elective surgery. *Br J Anaesth* 2019; **122**: 563–74
32. McEvoy MD, Gupta R, Koepke EJ, et al. Perioperative quality initiative consensus statement on postoperative blood pressure, risk and outcomes for elective surgery. *Br J Anaesth* 2019; **122**: 575–86
33. Khanna A, Pequeno P, Gupta S, et al. Increased risk of all cardiovascular disease subtypes among childhood cancer survivors: population-based matched cohort study. *Circulation* 2019; **140**: 1041–3
34. Ariyaratna D, Bhonsle A, Nim J, et al. Intraoperative vasopressor use and early postoperative acute kidney injury in elderly patients undergoing elective noncardiac surgery. *Ren Fail* 2022; **44**: 648–59

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